Response Letter with Annotations

Citation (APA format, using Zotero or other bibliography software)

Park, J. S., O'Brien, J. C., Cai, C. J., Morris, M. R., Liang, P., & Bernstein, M. S. (2023). Generative agents: Interactive simulacra of human behavior. In Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology. ACM. https://doi.org/10.1145/3586183.3606763

Annotations

1. **Purposes**
    - **a.** Motivation (one-sentence summary)

To build interactive agents that simulate believable human behavior over time by combining a large language model with memory, reflection, and planning mechanisms.

- **b.** Specific Rationale (several bullet points)

• They use a Sims-like sandbox world to demonstrate that such an architecture can produce both believable individual behavior and emergent group behavior, including information diffusion, relationship formation, and coordination.
• The authors therefore argue that believable agents require an architecture that can retrieve relevant past events, reflect on them to form higher-level inferences, and use those inferences to plan and react in ways that remain coherent across time.
• Although large language models can generate plausible behavior at a single time point, they do not by themselves maintain long-term coherence, manage growing memories, or support cascading social interactions across multiple agents.
• Prior work on believable agents relied mainly on rule-based scripting, reinforcement learning in narrowly specified environments, or cognitive architectures with manually authored procedures, all of which struggled to support open-world, socially coherent behavior over time.

- **c.** Research Questions and/or Hypotheses

The paper does not state formal hypotheses. Instead, it poses design- and evaluation-oriented research questions: (1) whether generative agents can produce believable individual behavior in narrowly defined contexts, (2) whether the major architectural components (observation/memory, reflection, and planning) causally contribute to believability, and (3) whether a community of agents can generate emergent social behaviors such as information diffusion, relationship formation, and coordination over time.

- Identify IV, DV, mediator, moderator

In the controlled evaluation, the main independent variable is **agent architecture condition**: the **full architecture**, **three ablated versions** that remove access to observation/memory, reflection, and/or planning, and a **human crowdworker-authored baseline**.
The main dependent variable is **perceived believability of the responses**, ranked by human evaluators. The paper does not test a mediator or moderator. In the end-to-end evaluation, the outcomes are descriptive rather than framed through IV-DV causal modeling; the authors track information diffusion, relationship formation, coordination, and errors.

- Identify specific relationships among variables

The controlled evaluation implies that **richer access to memory, reflection, and planning should increase behavioral believability**. The ablation logic specifically tests whether removing reflection and planning lowers believability relative to the full architecture. In the end-to-end simulation, the authors expect that important information will diffuse, that agents will form ties over time, and that agents will be able to coordinate around shared activities such as Isabella Rodriguez's Valentine's Day party.

2. Method

a. Research Design

The study combines system design with two empirical evaluations: (1) a controlled evaluation of individual agent behavior through structured natural-language interviews, and (2) an end-to-end deployment of 25 agents interacting continuously over two full game days in the Smallville sandbox world.

- Experiment:

A **within-subjects design** in which 100 human evaluators compared responses generated under five conditions for the same agent. The conditions were the full generative-agent architecture, three ablated architectures, and a human crowdworker-authored condition. For each participant, the study showed one randomly chosen interview question from each of five categories—self-knowledge, memory, plans, reactions, and reflections—and participants ranked the five responses from most to least believable.

The paper also includes a qualitative inductive analysis of response differences across conditions in the controlled evaluation, and an inductive analysis of boundary conditions and erratic behaviors observed during the end-to-end simulation.

b. Sample

For the controlled evaluation, the sample consisted of **100 adult English-speaking human evaluators** recruited from Prolific in the United States. This was not a representative population sample; it was a convenience sample used to evaluate believability rankings. The paper also used one crowdworker per agent to create the human-authored baseline responses. In the system deployment itself, the simulated community contained 25 unique agents in Smallville.

c. Measures

Key measures in the controlled evaluation were human rankings of response believability across the five conditions and qualitative distinctions in the responses. Key measures in the end-to-end evaluation were: (a) information diffusion, operationalized as whether each agent knew about Sam Moore's mayoral candidacy and Isabella Rodriguez's Valentine's Day party at the end of the simulation; (b) relationship formation, operationalized through mutual knowledge ties and summarized with network density; (c) coordination, operationalized as the number of invited agents who actually attended the party; and (d) observed errors and boundary conditions, including hallucinations and inappropriate action choices.

d. Analysis

The authors converted the rank-order believability data into interval-like condition scores using TrueSkill ratings. They also used a Kruskal-Wallis test to assess overall rank differences across conditions, followed by Dunn post-hoc tests with Holm-Bonferroni correction for pairwise comparisons. In addition, the first author conducted qualitative open coding in two phases to identify response patterns. For the end-to-end simulation, the analysis was descriptive and interview-based: the authors labeled whether agents knew particular pieces of information, constructed an undirected relationship graph from mutual knowledge reports, calculated network density, counted party attendance, and then performed an inductive analysis of common failures.

3. Results and Conclusions

a. Answers to RQs and Hypotheses

The controlled evaluation shows that the full architecture produced the most believable behavior, outperforming all ablated conditions and the human crowdworker baseline. This supports the authors' claim that observation/memory, reflection, and planning each contribute to believable behavior. The end-to-end evaluation shows that generative agents can produce emergent social outcomes: information diffused through the community, new ties formed, and some invited agents coordinated successfully around the Valentine's Day party.

b. Include Key Figures/Tables

Key figures include Figure 5, which presents the overall architecture linking perceive, memory stream, retrieve, reflect, plan, and act;

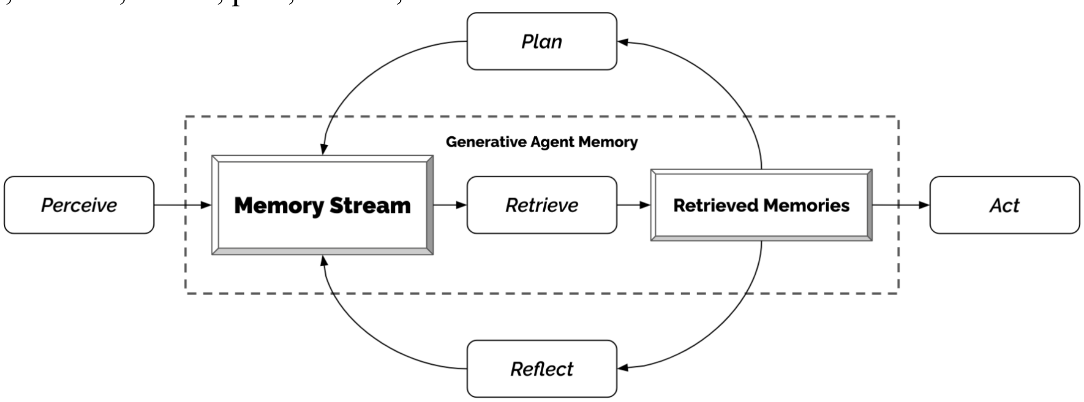

Figure 5: Our generative agent architecture. Agents perceive their environment, and all perceptions are saved in a comprehensive record of the agent's experiences called the memory stream. Based on their perceptions, the architecture retrieves relevant memories and uses those retrieved actions to determine an action. These retrieved memories are also used to form longer-term plans and create higher-level reflections, both of which are entered into the memory stream for future use.

Figure 6, which illustrates memory retrieval based on recency, importance, and relevance;

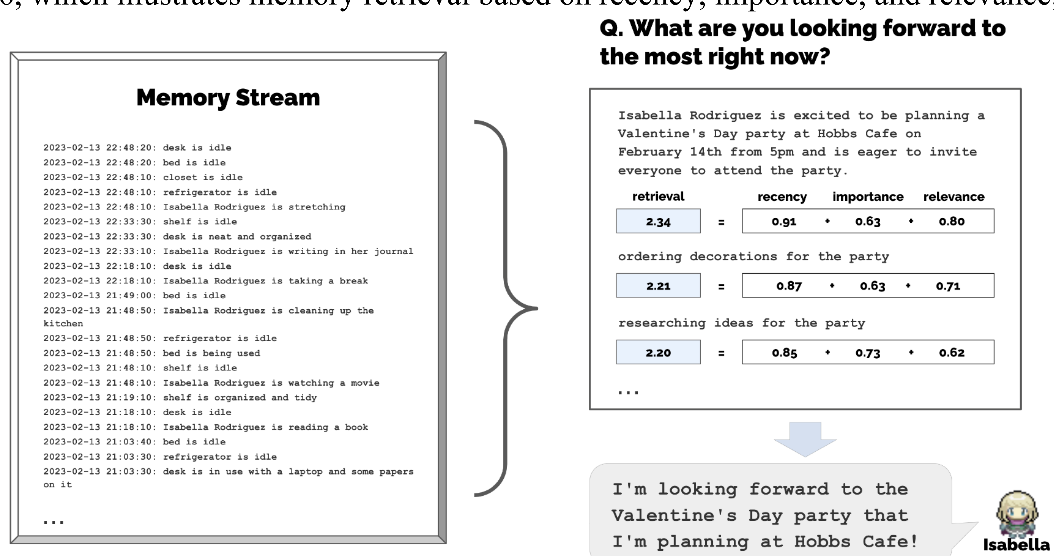

Figure 6: The memory stream comprises a large number of observations that are relevant and irrelevant to the agent's current situation. Retrieval identifies a subset of these observations that should be passed to the language model to condition its response to the situation.

Figure 7, which shows the reflection tree for Klaus Mueller;

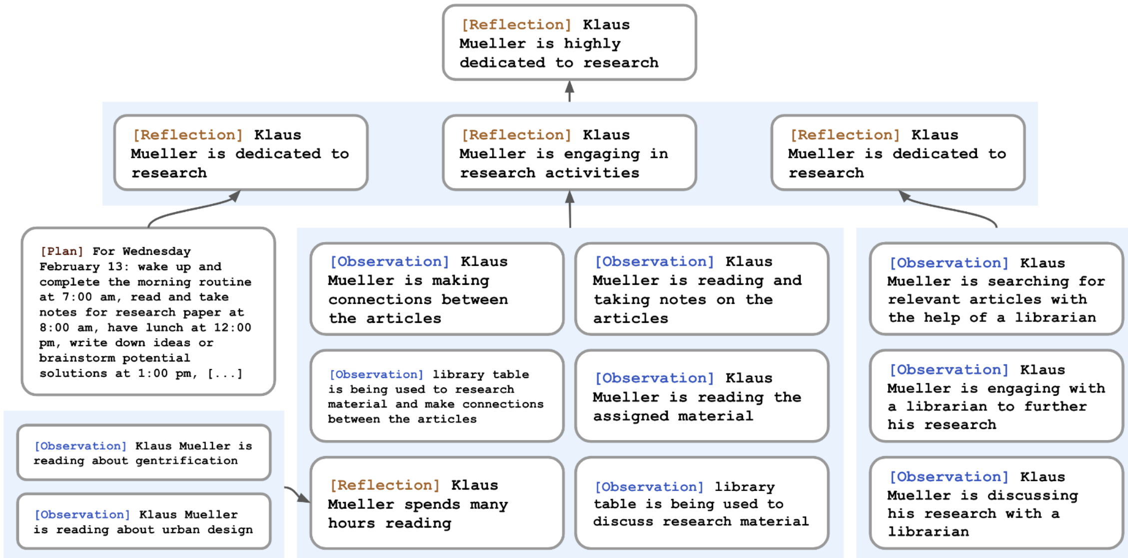

Figure 7: A reflection tree for Klaus Mueller. The agent's observations of the world, represented in the leaf nodes, are recursively synthesized to derive Klaus's self-notion that he highly dedicated to his research.

Figure 8, which reports the controlled-evaluation result that the full architecture was most

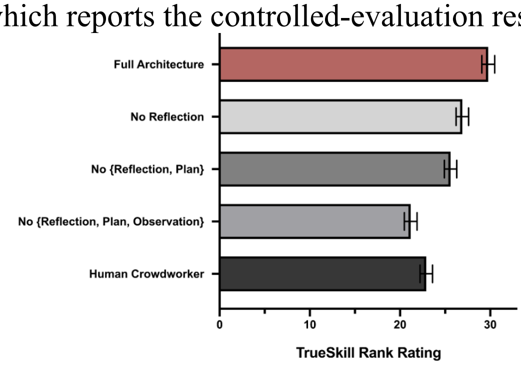

Figure 8: The full generative agent architecture produces more believable behavior than the ablated architectures and the human crowdworkers. Each additional ablation reduces the performance of the architecture.

believable;

and Figure 9, which visualizes the diffusion path of Isabella Rodriguez's Valentine's Day party invitation. Numerically, the full architecture received the highest TrueSkill score (mu = 29.89, sigma = 0.72), followed by the no-reflection condition (mu = 26.88, sigma = 0.69), the no-reflection/no-planning condition (mu = 25.64, sigma = 0.68), the crowdworker condition (mu = 22.95, sigma = 0.69), and the fully ablated baseline (mu = 21.21, sigma = 0.70). In the two-day simulation, knowledge of Sam's candidacy increased from 1 agent to 8 agents, knowledge of Isabella's party increased from 1 agent to 13 agents, network density increased from 0.167 to 0.74, and 5 of 12 invited agents attended the party.

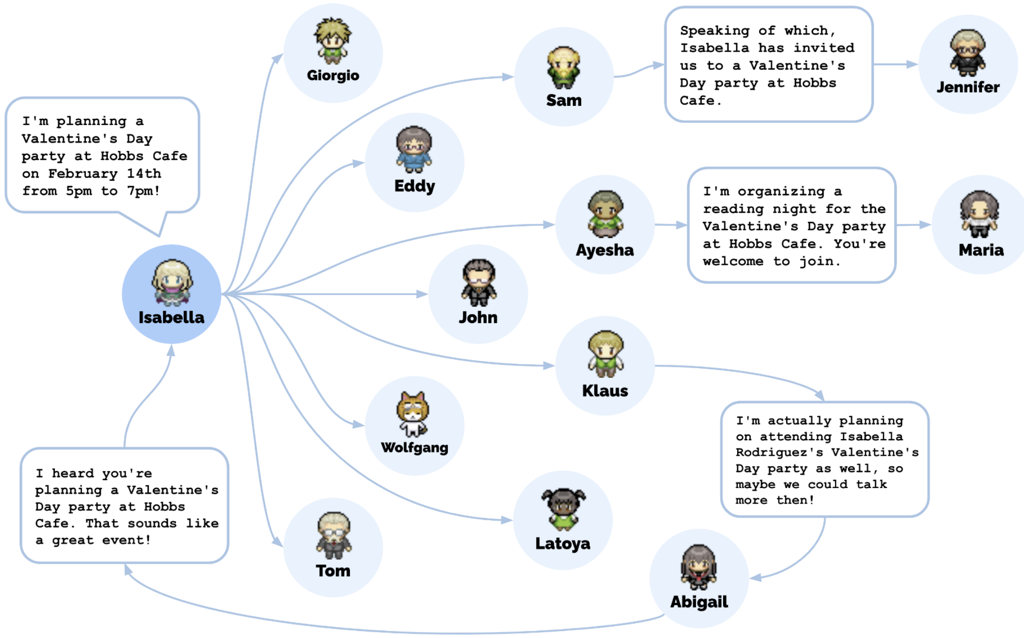

Figure 9: The diffusion path for Isabella Rodriguez's Valentine's Day party invitation involved a total of 12 agents, aside from Isabella, who heard about the party at Hobbs Cafe by the end of the simulation.

c. Authors' Conclusions

The authors conclude that generative agents can serve as believable simulacra of human behavior when a large language model is supplemented with mechanisms for memory, reflection, and planning. They argue that the architecture enables both believable individual behavior and emergent group dynamics in interactive environments. At the same time, they acknowledge that the architecture still fails at times through imperfect retrieval, embellishment, hallucination, formalized dialogue style, and other inherited limitations of the underlying language model.

4. Discussion

a. Implications

The paper suggests that generative agents could be useful for role-play, social prototyping, virtual worlds, games, immersive environments, social robotics, and human-centered design. More broadly, the work implies that agent architectures grounded in long-term memory and reflection can make LLM-based systems more useful for simulating dynamic human behavior rather than only producing one-off responses.

b. Limitations

The authors identify several limitations: imperfect retrieval of relevant memories; hallucinated embellishments; inappropriate or less believable action choices as agents learn more locations; misclassification of environmental norms such as bathroom occupancy or store closing times; overly formal and overly cooperative dialogue likely caused by instruction tuning; high computational cost and long runtime; short evaluation timescale; lack of a maximal-human-performance baseline; limited knowledge about robustness to prompt hacking, memory hacking, and hallucination; and the possibility that biases in the underlying large language model may produce biased or stereotypical behavior.

c. Future Directions

Future work could improve the retrieval module by better tuning the relevance, recency, and importance functions; reduce cost and increase real-time interactivity through parallelization or agent-specific models; extend evaluation to longer timescales and stronger human benchmarks; compare different underlying language models and hyperparameters; and test robustness more systematically against prompt hacking, memory hacking, and hallucination. The authors also suggest that better alignment of the underlying language model will be necessary to address deeper issues of bias and behavior quality.

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Response Letter (~600 words)

- Your assessment of the study

This study is compelling because it does not simply claim that large language models can imitate humans; instead, it shows that believable long-term behavior requires architectural support around the language model. The paper is strongest when it explains why first-order prompting is insufficient and then operationalizes that argument through memory, reflection, and planning. The architecture is conceptually clear, the Smallville demonstration is easy to understand, and the two-part evaluation gives the paper both a controlled test of the components and a broader demonstration of emergent social behavior. I also think the ablation design is especially effective because it turns a broad systems claim into a more credible causal argument: the paper does not merely show that the full system works, but that removing reflection and planning reduces perceived believability. Figure 8 makes that point especially clearly. The paper is also unusually careful for a systems paper in that it does not hide its boundary conditions. It openly reports retrieval failures, hallucinated embellishments, and the influence of instruction tuning on dialogue style and cooperativeness.

- Weaknesses

The main weakness is that the paper's central outcome—believability—is still a subjective judgment rather than a behavioral or externally validated outcome. The controlled evaluation is well designed for what it is trying to show, but the use of a convenience sample of Prolific raters and a relatively weak crowdworker baseline limits how strongly we should interpret the results. The end-to-end evaluation is also more descriptive than rigorous: the authors document information diffusion, relationship formation, and coordination, but these are not benchmarked against human communities or against alternative computational systems. In addition, several measures rely on agent self-reports in interviews, even though the paper itself shows that agents can retrieve incomplete memories and embellish what they know. The paper does attempt to verify some of these responses against the memory stream, but the possibility of distorted self-report remains important. Finally, the architecture is expensive and slow, and the agents remain dependent on the strengths and biases of the underlying language model, which means that some of the most serious limits are not resolved at the architectural level.

- Other notes

One point I found especially important is that the paper should not be read as claiming true human-like agency. The authors explicitly state that the behaviors are intended as believable simulacra rather than genuine agency, and that distinction matters conceptually. I also think the ethical discussion is not ornamental here; it is directly connected to the paper's success. The more believable these agents become, the more serious the risks of parasocial attachment, persuasion, and over-reliance become. More generally, this paper seems most valuable as a foundational systems paper: it proposes a reusable architecture and demonstrates a direction for future work rather than delivering a final solution to human simulation. Its importance lies less in proving that the agents are fully realistic and more in showing what kinds of architectural components are necessary if researchers want LLM-based agents to behave coherently over time.